Exploring African Talent in Europe - A Data Management Project

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Abstract

This project addresses the challenges football teams face during player scouting and acquisition. By developing a comprehensive data management system focused specifically on African footballers competing in Europe's top five leagues, the project aims to optimize the recruitment process. The research will collect, organize, store, integrate and analyze extensive performance data on players from the top 15 African football nations, creating detailed profiles based on position metrics, playing styles, physical attributes, and historical performance indicators. This approach will enable teams to identify talent that matches their tactical requirements and organizational needs precisely. The ultimate goal is to create a more efficient way for clubs to discover and acquire African talents that align with their strategic objectives and team culture, thus improving both sporting performance and financial sustainability.

1 Introduction

The project aims to contrast the challenges faced by football teams during the scouting and acquisition process of players. Scouting can often be a time-consuming, costly and complex task, which requires teams to assess various attributes and characteristics before making a purchase decision. This project seeks to simplify this process by providing data-driven insights that help teams in identify the right African football players who meet specific needs and criteria. Using data management techniques, the project will collect and analyze comprehensive data on African footballers, focusing on key performance indicators, skill sets, and other relevant factors that influence player selection. The goal is to help football teams make more informed and efficient decisions by providing them with detailed and accessible player profiles based on characteristics such as position, playing style, physical attributes, and historical performance. Lastly, this project aims to simplify and enhance the player acquisition process, reducing both the time and cost typically involved in scouting, while ensuring that teams are able to identify players who align with their strategic goals and team dynamics.

2 Research goal and Data understanding

2.1 Research goal

The research will collect comprehensive performance metrics, creating detailed player profiles that encapsulate position skills, playing styles, physical measurements, and historical statistical performance. This systematic approach will allow clubs to move beyond subjective assessments and utilize objective data to match player capabilities with team requirements. Furthermore, by focusing specifically on talents from the top 15 African football nations, the project addresses a segment of players who are often undervalued in the transfer market despite their significant potential.

2.2 Data understanding

Identification and understanding of the data are crucial steps in avoiding problems in the next stages of analysis. We focus on football players who play for clubs in the top five European leagues and hold a nationality ranked among the top 15 African football nations. The list of national teams to consider was obtained from the fifa website [FIF25a].

	Position	Country	Points
1	14°	Morocco	1688.18
2	17°	Senegal	1637.25
3	33°	Egypt	1513.48
4	37°	Algeria	1495.85
5	44°	Nigeria	1482.23
6	46°	Ivory Coast	1478.47
7	49°	Cameroon	1471.49
8	51°	Mali	1466.10
9	52°	Tunisia	1464.42
10	57°	South Africa	1428.87
11	61°	Democratic Republic of Congo	1395.77
12	66°	Burkina Faso	1373.76
13	72°	Capo Verde	1328.51
14	77°	Ghana	1320.52
15	78°	Guinea	1311.88

Table 1: FIFA rank for top 15 African countries at December 19, 2024

From Table (1) The highest-ranked team is Morocco, the top African team (14th in the overall FIFA ranking) with 1688.18 points. Senegal follows with 1637.25 points (17th globally). And in third place is Egypt with 1513.48 points (33rd worldwide). The ranking points are calculated using the SUM formula [FIF25b], introduced in August 2018. This formula considers match results, opponent strength, competition importance, and home/away factors. The formula is as follows:

$$P = P_{\text{prev}} + I \cdot (W - W_{\text{exp}})$$

where:

- P is the updated team points.
- P_{prev} is the previous team points.
- I is the importance factor of the match.
- W is the actual result of the match (1 for a win, 0.5 for a draw, 0 for a loss).
- W_{exp} is the expected match result.

Among all the available sources to accomplish our aim, we identify two fundamental sources that provide football players data, Sofifa [Sof24] and Transfermakt [Tra24].
You may wonder why we chose to perform our analysis according to Sofifa ratings. The first source Sofifa, which is based on data provided by EA SPORTS, was chosen for the following main reasons:

- **Game Performance**: collects real-time data from live matches, which is generated from major football leagues, tournaments, and competitions worldwide. Data are gathered on individual player actions, such as goals scored, assists, tackles, shots on target, dribbles attempted, successful passes, and so on.
- **Physical Performance**: attributes such as speed, acceleration, and distance covered; they rely on technologies such as GPS trackers. Some top-level teams use GPS trackers, heart rate monitors, and other wearable devices to monitor player performance during games and training sessions. These devices track movement, speed, distance covered, acceleration, and deceleration, all of which contribute to a player's speed and other athletic attributes.
- **Video Analysis**: uses video analysis tools, like Wyscout [Wys24] or Hudl [Hud24], that break down footage from games, enabling analysts to measure things like dribbles, shots, headers, and defensive actions with high precision.
- **Other Sources**: Companies such as Opta Sports [Opt24], StatsBomb [Sta24], or InStat are frequently used by analytics platforms to gather event-level data, providing detailed breakdowns of each player's actions during a match. Many football clubs and analytical platforms, including EA Sports, also rely on

scouts and expert analysts who provide subjective insight into a player's technical and physical abilities. Players who have consistently performed well over multiple seasons are often valued higher than those with inconsistent performances.

- **Injuries**: Players who have had fewer injuries tend to have higher values, as they are more likely to maintain their form and play consistently.
- **Fans' Favorites**: A player who is popular among fans might be valued higher due to their commercial potential, even if their on-field performance doesn't fully match their profile.

The features regarding players that we considered from Sofifa are the following:

- **Name**: the player's full name.
- **Age**: the player's current age. This is important because it impacts both their potential for growth and the stage of their career.
- **Potential**: indicates the player's maximum possible rating or ability in their career, reflecting how much they can improve based on their current age and development.
- **Team**: this refers to the team the player is currently playing for.
- **Contract**: shows details about the player's contract (e.g., length, expiration date, etc.).
- **Height**: the player's height in centimeters or feet/inches. This can influence how they perform in aerial duels, their movement, and their overall physical profile.
- **Weight**: The player's weight in kilograms or pounds, which also impacts their physicality, balance, and stamina.
- **Preferred Foot**: this is the player's dominant foot (left or right). It's important for determining how well they perform tasks like shooting, passing, or dribbling.
- **Growth**: this refers to the player's ability to improve in terms of skill and ability over time, often tied to their age and potential.
- **Value**: the market value of the player, which is a reflection of their ability, age, and how much they could potentially be sold for in the transfer market.
- **Wage**: the player's weekly salary or wage, which often reflects their value to the team and their experience or skill level.
- **Crossing**: the ability to accurately deliver the ball across the pitch to teammates, often in attacking situations.
- **Finishing**: a player's ability to convert scoring chances into goals, especially in one-on-one situations.
- **Heading Accuracy**: how well a player can direct headers, whether offensively or defensively.
- **Short Passing**: the accuracy of passes made over short distances.
- **Volleys**: a player's ability to strike a ball cleanly in mid-air, especially in attacking positions.
- **Dribbling**: how well a player can control and maneuver the ball past opponents.
- **Curve**: the ability to apply a curve or bend on the ball, useful for free kicks and passes.
- **FK Accuracy**: the precision with which a player can strike free kicks.
- **Long Passing**: a player's ability to accurately pass the ball over longer distances.
- **Ball Control**: the ability to retain the ball under pressure.
- **Acceleration**: how quickly a player can reach top speed from a stationary position.
- **Sprint Speed**: the maximum speed a player can reach when running at full pace.

- **Agility**: how quickly a player can change direction and react to in-game movements.
- **Reactions**: the speed at which a player can respond to actions on the pitch (e.g., reacting to a loose ball).
- **Balance**: a player's ability to maintain stability while on the ball or in physical duels.
- **Shot Power**: the strength behind a player's shot, impacting how hard they can strike the ball.
- **Jumping**: the ability to leap effectively, useful for headers and aerial duels.
- **Stamina**: the endurance to perform at a high level throughout a match.
- **Strength**: a player's physicality and ability to resist challenges from opponents.
- **Long Shots**: the ability to score goals from outside the penalty box, requiring both accuracy and power.
- **Aggression**: a player's willingness to press, challenge, and challenge for the ball.
- **Interceptions**: the ability to read the game and cut out passes by positioning oneself effectively.
- **Vision**: a player's ability to spot passes and make creative decisions on the field.
- **Penalties**: the player's proficiency in converting penalty kicks.
- **Composure**: a player's mental toughness and ability to remain calm in high-pressure situations, especially when finishing or defending.
- Defensive Awareness: how well a player can read and anticipate defensive situations.
- **Standing Tackle**: the ability to tackle an opponent without committing to a risky slide, using good body positioning and timing.
- **Sliding Tackle**: the ability to win the ball from an opponent by sliding to the ground, often in defensive situations.
- **GK Diving**: a goalkeeper's ability to dive and reach shots outside of their body.
- **GK Handling**: the goalkeeper's ability to catch, block, or control the ball.
- **GK Kicking**: the accuracy and strength of a goalkeeper's kicks, including goal kicks and distribution.
- **GK Positioning**: the goalkeeper's awareness of where to position themselves in relation to the ball and goal.
- **GK Reflexes**: how quickly a goalkeeper can react to fast or unexpected shots.
- **Weak Foot**: a player's ability to use their non-dominant foot effectively for passing, dribbling, or shooting.
- **Skill Moves**: the player's ability to perform complex dribbling skills and tricks.

Transfermarkt is one of the largest databases of football players in the world. Covers a wide range of leagues, competitions, and player profiles. This extensive reach ensures that their valuations are based on a broad and diverse dataset. One of Transfermarkt's strongest features is its detailed tracking of transfer transactions, including transfer fees, contract lengths, and transfer history. By analyzing the history of player transfers and the amounts involved, it calculates the trends and patterns in player market values. Transfermarkt includes details of the player's contract, such as its start and end dates, crucial to determining a player's current market value. Players nearing the end of their contracts may see a drop in their value due to the possibility of a free transfer. The following are the information obtained from Tranfermakt:

- **League**: the competition or tournament in which a player competes. HIt has five levels, which are Premier League (EPL), La Liga (Spain), Serie A (Italy), Bundesliga (Germany), and Ligue 1 (France).
- **Season**: the specific year or time period. A season typically runs for about a year, starting in the middle of the summer. For example, "2024–2025" would refer to the competition that takes place from summer 2024 to spring 2025.

- **Country**: the country from which the player comes.
- **Player Name**: this is the name of the player to whom the player is referred.
- **Position**: refers to the specific role or position the player plays on the team.
- **Club**: the football team that the player represents.
- **Debut Date**: the date on which the player first appeared in a competitive match for their club or national team, marking their official entry into professional or international football.
- **Appearances**: refers to the number of times a player has played in matches (starting or coming on as a substitute) of the league for their club during a specific season.
- **Goals**: refers to the number of goals that the player has scored in competitive league matches for their club.

3 Data acquisition (scraping)

Since both sources do not provide a way to download or extract the data directly, we use scraping to automate the data extraction process. In particular, selenium on Sofifa to simulate the code useful to transform the website into HTML and extract the data, and BeautifulSoup on Transfermarket to extract the webpage for our analysis.

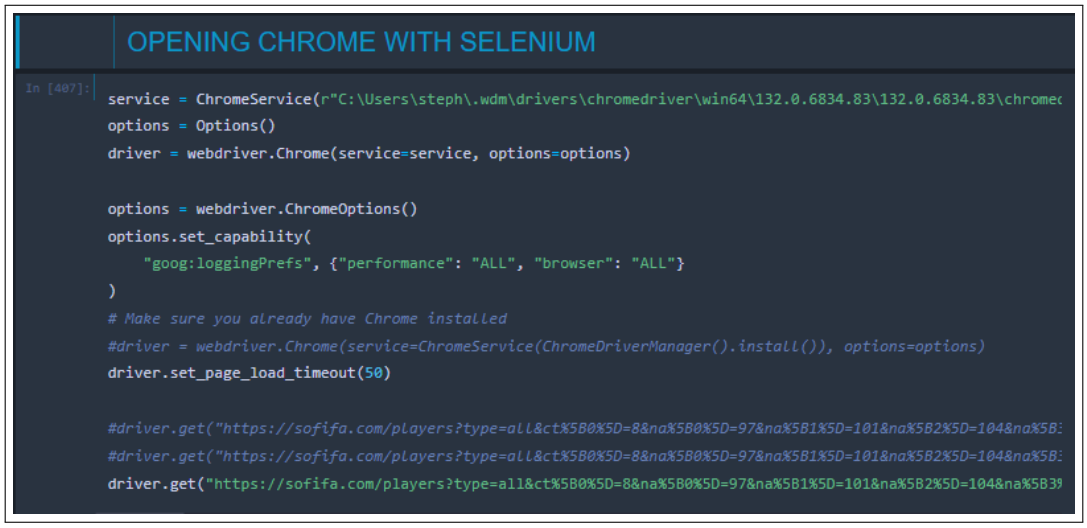
3.1 Selenium on Sofifa

If the page is not in HTML but the server sends the code to the page to create the HTML page, in order to scrape we have to simulate the request of the code. The solution in this case is Selenium, it is a library for controlling web browsers through programs and performing browser automation. Selenium is a powerful tool for automating web browsers. It is especially useful for scraping dynamic content (content that changes after the initial page load). We set up a Selenium WebDriver (e.g. ChromeDriver, FirefoxDriver) to automate browser interactions and to navigate through web pages. This can involve clicking buttons (NEXT), filling out forms, scrolling, and waiting for dynamic content to load. we used Selenium's methods like find elements by class name(), to locate and extract the table that we were interested in.

```
In [4]: import json,requests
        from selenium import webdriver
        from selenium.webdriver.chrome.service import Service as ChromeService
        from selenium.webdriver.chrome.options import Options
        from selenium.webdriver.common.desired_capabilities import DesiredCapabilities
        from selenium.webdriver.common.by import By
        from selenium.webdriver.support.ui import WebDriverWait
        from selenium.webdriver.support import expected_conditions as EC
        import pandas as pd
        import time
        from webdriver_manager.chrome import ChromeDriverManager
```

Figure 1: Libraries used to scrape data from Sofifa through Selenium

From Figure (1) Selenium is used to interact and automate a Chrome browser. Whilst import 'JSON,requests' is used for sending HTTP. Pandas is used to store and manipulate scraped data in tabular form. time provides time-related functions. It is commonly used for adding delays or pauses in the code (like 'time.sleep()'), which can be useful for ensuring that pages have fully loaded before interacting with them. WebDriverManager automatically handles the installation and management of the correct ChromeDriver.



OPENING CHROME WITH SELENIUM

```
In [407]:
service = ChromeService(r"C:\Users\steph\.wdm\drivers\chromedriver\win64\132.0.6834.83\132.0.6834.83\chromec
options = Options()
driver = webdriver.Chrome(service=service, options=options)

options = webdriver.ChromeOptions()
options.set_capability(
    "goog:loggingPrefs", {"performance": "ALL", "browser": "ALL"}
)
# Make sure you already have Chrome installed
#driver = webdriver.Chrome(service=ChromeService(ChromeDriverManager().install()), options=options)
driver.set_page_load_timeout(50)

#driver.get("https://sofifa.com/players?type=all&ct%5B0%5D=8&na%5B0%5D=97&na%5B1%5D=101&na%5B2%5D=104&na%5B3
#driver.get("https://sofifa.com/players?type=all&ct%5B0%5D=8&na%5B0%5D=97&na%5B1%5D=101&na%5B2%5D=104&na%5B3
driver.get("https://sofifa.com/players?type=all&ct%5B0%5D=8&na%5B0%5D=97&na%5B1%5D=101&na%5B2%5D=104&na%5B39
```

Figure 2: Opening Chrome with Selenium

Figure (2) explained:

- 'service' line sets up the path to the Chrome WebDriver executable. The chromedriver.exe file is the bridge between Selenium and the Chrome browser, enabling us to control Chrome via Selenium.
- 'options = Options()' creates a new Options object that allows one to configure different settings for the Chrome browser.
- 'driver = webdriver.Chrome(service=service, options=options)' initializes the ChromeDriver using the service and options settings, launching the Chrome browser.
- 'options = webdriver.ChromeOptions()' creates a new ChromeOptions object.
- 'set_capability' method is used to enable logging capabilities for Chrome. This configuration sets up logging for both performance and browser logs. It is useful for debugging or analyzing the behavior of the browser during web scraping.
- 'driver.set_page_load_timeout(50)' sets the page load timeout to 50 seconds. Prevents the program from waiting indefinitely for a page to load.
- The last line opens the specified URL in the Chrome browser. The URL links to a page on Sofifa that displays a list of football players with various attributes. The driver.get() method makes the browser visit the provided URL.

configuring the driver to scrape the table from the page for each of the three seasons and clicking the NEXT buttom if there is another page

In [167]:

```
# Configure driver with timeout
driver.set_page_load_timeout(25)  # wait 25 seconds for the page to load

all_data = []

try:
    while True:
        # Wait till the table is loaded
        WebDriverWait(driver, 20).until(EC.presence_of_element_located((By.TAG_NAME, 'table')))

        # Find table an rows
        table = driver.find_element(By.TAG_NAME, 'table')
        rows = table.find_elements(By.TAG_NAME, 'tr')

        for row in rows:
            cells = row.find_elements(By.TAG_NAME, 'td')
            cells_text = [cell.text.strip() for cell in cells]
            if cells_text:
                all_data.append(cells_text)

        # Scroll to the 'NEXT' button
        driver.execute_script("window.scrollTo(0, document.body.scrollHeight);")
        time.sleep(2)

        # Select "Next" and not "Previous"
        try:
            pagination = driver.find_element(By.CLASS_NAME, "pagination")
            next_buttons = pagination.find_elements(By.CLASS_NAME, "button")

            # Finds 'Next' in the elements
            next_button = None
            for btn in next_buttons:
                if "Next" in btn.text:
                    next_button = btn
                    break
```

Figure 3: Scrape the table from the page for each of the three seasons pt.1

Figure (3) script extracts player data from tables across multiple pages, handling pagination by clicking the ”Next” button to advance through the pages.

- With ’driver.set_page_load_timeout(25)’ we configure the WebDriver with a time to load before timeout, of 25 seconds.
- An empty list ’all_data = []’ is then created to store the scraped data.
- Inside the try block with a while True loop we wait for a table element to appear on the page using WebDriverWait.
- Locate the table element using driver.find_element(By.TAG_NAME, ’table’).
- Find all rows in the table with table.find_elements(By.TAG_NAME, ’tr’).
- For each row found in the table, we find all cells with row.find_elements(By.TAG_NAME, ’td’).
- Extract the text from each cell and strip whitespace with [cell.text.strip() for cell in cells], if there is text content in the cells, it appends it to the all_data list.
- After scraping the table, we Scroll to the bottom of the page and wait for 1 second with time.sleep(1).

- Then find and click the ”Next” button.
- Locate a pagination element by the class name ”pagination”.
- Find all button elements within pagination.
- Search through these buttons to find one with the text ”Next”.
- If found, it will presumably click this button (though the click action is not shown in this snippet).

```
                if next_button is None:
                    print("Nessun pulsante 'Next' trovato. Fine delle pagine.")
                    break

            except:
                print(" Nessun pulsante 'Next' trovato. Fine delle pagine.")
                break

            # Scroll till the button to make it visible
            driver.execute_script("arguments[0].scrollIntoView();", next_button)
            time.sleep(1)

            # Use JavaScript to click in case there's any issue with clicking
            driver.execute_script("arguments[0].click();", next_button)

            # Wait till the new page loads the data
            WebDriverWait(driver, 20).until(EC.staleness_of(table))

        # Convert data into a dataframe
        df = pd.DataFrame(all_data)

        # Save to a csv file
        df.to_csv('all_players_2024.csv', index=False, encoding='utf-8')
        print("Dati di tutte le pagine salvati in all_players_2023.csv")

    finally:
        driver.quit()
```

Figure 4: Scrape the table from the page for each of the three seasons pt.2

- If next_button is None, it prints ”No ’Next’ button found”. End of page. Then it breaks out of the loop using break.
- If an error occurs when trying to find the ”Next” button, it prints the same message and breaks out of the loop.
- If a ”Next” button is found, the script proceeds to scroll the button into view using JavaScript (to bypass any issue that may occur) with the line: driver.execute_script(”arguments[0].scrollIntoView();”, next_button).
- Pause briefly with time.sleep(1) to allow the scrolling to complete.
- Use JavaScript to click the button: driver.execute_script(”arguments[0].click();”, next_button)
- This dual approach (scrolling first, then clicking) helps ensure the button is visible before attempting to interact with it.
- After clicking the button we waits for the new page to load using WebDriverWait(driver, 20).until(EC.staleness_of(table)), indicates that the page has refreshed.

- Once all pages have been processed and the loop is stopped (when no ”Next” button is found), the collected data is converted into a pandas DataFrame: df = pd.DataFrame(all_data) and it is saved to a CSV file.
- Finally, the WebDriver is closed with driver.quit()

	Name	Age	Potential	Team & Contract	Height	Weight	Prefered foot	Growth	value	Wage	...	Standing tackle	Sliding tackle	GK diving
0	M. Salah\nRW	29	91	Liverpool\n2017 ~ 2023	175cm / 5'9"	71kg / 157lbs	Left	0	€129M	€350K	...	43	41	14
1	S. Mané\nLW CF	29	89	Liverpool\n2016 ~ 2023	175cm / 5'9"	69kg / 152lbs	Right	0	€101M	€270K	...	42	38	10
2	É. Mendy\nGK	29	87	Chelsea\n2020 ~ 2025	197cm / 6'6"	86kg / 190lbs	Right	1	€52M	€140K	...	11	13	85
3	R. Mahrez\nRW RM	30	86	Manchester City\n2018 ~ 2025	179cm / 5'10"	67kg / 148lbs	Left	0	€65.5M	€220K	...	31	22	15
4	K. Koulibaly\nCB	30	86	Napoli\n2014 ~ 2023	187cm / 6'2"	89kg / 196lbs	Right	0	€55.5M	€105K	...	89	86	7
5	A. Hakimi\nRB RWB	22	89	Paris Saint-Germain\n2021 ~ 2026	181cm / 5'11"	73kg / 161lbs	Right	4	€73M	€100K	...	77	76	10
6	T. Partey\nCDM CM	28	84	Arsenal\n2020 ~ 2024	185cm / 6'1"	77kg / 170lbs	Right	0	€37M	€115K	...	84	80	11
7	J. Matip\nCB	29	84	Liverpool\n2016 ~ 2024	194cm / 6'4"	90kg / 198lbs	Right	0	€35.5M	€145K	...	88	82	11

Figure 5: A portion of the dataframe after scraping using Selenium

3.2 BeautifulSoup on Transfermarkt

Beautiful Soup is a popular Python library used for web scraping purposes. It makes it easy to extract data from web pages and process them in a format that can be used for further analysis or manipulation. When scraping data from the Web, websites usually return raw HTML or XML content. Beautiful Soup helps you parse this content and navigate the structure of the page to find the data you need (such as tables, paragraphs, links, or images). Provides simple methods to search for and extract information from HTML elements based on tags, attributes, and classes.

Libraries

```
import pandas as pd
import requests
from bs4 import BeautifulSoup
from unidecode import unidecode
from datetime import datetime
import matplotlib.pyplot as plt
import seaborn as sns
import numpy as np
```

Figure 6: List of libraries used scraping with Beautifulsoup

From Figure 6, we can see the list of libraries that were used to scrape Transfermarkt webpage. requests is used for making HTTP requests. BeautifulSoup (from bs4) is a library for parsing HTML and XML documents, often used for web scraping. unidecode helps convert Unicode text into an ASCII representation. datetime (from datetime) provides functions for working with dates and times. matplotlib.pyplot (plt) is a plotting library for creating graphs and visualizations. seaborn (sns) is built on matplotlib, it provides advanced statistical data visualization. numpy (np) is a library for numerical computing, supporting arrays and mathematical operations.

To scrape the data, we first create three dictionaries. The first one named leagues contains the Tranfermarkt

ids of the top 5 european leagues (GB1, FR1, L1, IT1 AND ES1), the second, seasons, has the three seasons of interst (2021, 2022, 2023), and lastly, 'countries' which has the 15 african countries of interest. The IDs will help us navigate the Transfermakt webpage to extract data. We include this header 'headers = "User-Agent": "Mozilla/5.0"' to avoid being blocked or flagged by the server as a bot, since some websites check the "User-Agent" string to identify automated traffic and block it. This would make the request appear more like it is coming from a legitimate browser rather than a script. At this point, we create the function 'scrape_table(url)' to scrape the football player data from the Transfermakt paginated table. Collect data on each player (name, position, club, debut date, appearances, and goals), handle pagination, and return the collected data as a list of player attributes.

```
headers = {"User-Agent": "Mozilla/5.0"}
# Function to scrape a table for a given URL
all_players = []

def scrape_table(url):
    players = []
    while url:
        # Request the page
        response = requests.get(url, headers=headers)
        soup = BeautifulSoup(response.content, "html.parser")
        # Find the table
        table = soup.find("div", class_="responsive-table")
        if not table:
            break
        rows = table.find_all("tr", class_=["odd", "even"])
        for row in rows:
            columns = row.find_all("td")
            if len(columns) >= 5:
                # Extract player name
                player_name = columns[2].text.strip()
                # Extract position
                position = columns[3].text.strip()
                # Extract club name
                club = club = columns[4].find("a")["title"].strip() if columns[4].find("a") and "title" in c
                # Extract and format debut date
                debut_date_raw = columns[5].text.strip()
                debut_date = datetime.strptime(debut_date_raw, "%b %d, %Y").strftime("%d/%m/%Y") if debut_da
                # Extract appearances
                appearances = columns[6].text.strip()
                # Extract goals
                goals = columns[7].text.strip()
                players.append([player_name, position, club, debut_date, appearances, goals])
        # Check for a "next" button for pagination
        next_page = soup.find("a", class_="tm-pagination__link--next")
        if next_page and "href" in next_page.attrs:
            url = "https://www.transfermarkt.co.uk" + next_page["href"]
        else:
            url = None

    return players
```

Figure 7: Function to scrape table with player data

Here, follow the code explanation in detail:

- all_players = []: this is an empty list to store all player data.
- Within the function, we start with players = [], a list that stores the data of the players on each page.
- While url: loop keeps running as long as there is a valid URL,a next page to scrape.
- Inside this loop, the page is requested using 'requests.get(url, headers=headers)'.

- The 'BeautifulSoup(response.content, "html.parser")' is used to parse the HTML content of the page.
- To extract data from the table 'table = soup.find("div", class_="responsive-table")' finds the table containing player data by looking for the ' ' tag with the class "responsive-table".
- if not table is found 'break' ensures the function stops scraping (breaks the loop).
- 'rows = table.find_all("tr", class_=["odd", "even"])' finds all the rows ('<tr>') of the table with class odd or even (individual palre rows).
- For each row, the player's information is extracted and appended as a list to the players' list.
- The 'next_page' object checks if there is a 'next' button (or pagination link) on the page.
- 'next_page = soup.find("a", class_="tm-pagination__link–next")' finds the 'next' button for pagination.
- If the 'next' button is found and contains a valid link (href), it appends the base URL ("https://www.transfermarkt.co.uk") to the relative link and assigns it to the URL.
- If there is no next page (i.e. the last page has been reached), it sets 'url = None', which breaks the loop and stops scraping.
- Once all pages are scraped and there are no more pages to scrape, the function returns the players list.

```
# Loop through leagues, seasons and countries to scrape data
for league, league_id in leagues.items():
    for season, season_id in seasons.items():
        for country, land_id in countries.items():
            print(f"Scraping data for {country} in {league} ({season})...")
            url = f"https://www.transfermarkt.co.uk/{league}/gastarbeiterdetail/wettbewerb/{league_id}?saisc
            country_data = scrape_table(url)
            for player in country_data:
                all_players.append([league, season, country] + player)
```

Figure 8: Loop through leagues, seasons and countries to scrape data

From Figure (8):

- The first loop to iterate through the leagues dictionary, where league is the name of the league (e.g. 'Premier League') and 'league_id' is the unique identifier for that league (e.g. 'GB1').
- The second loop to iterate through the seasons dictionary, where season is the name of the season (e.g., '2021/2022') and 'season_id' is the unique identifier for that season (e.g. '2021').
- The third loop iterates through the countries dictionary, where country is the name of the country (e.g. 'Ghana') and 'land_id' is the unique identifier for that country (e.g. '54').
- The 'url' variable generates a URL to access player data on the Transfermarkt website, it includes the league, season, and country IDs to request data for the specific combination.
- 'country_data = scrape_table(url)' calls the 'scrape_table' function to scrape data from the generated URL. The 'scrape_table' function stores in 'country_data' a list of players and their relevant information for the given country, league, and season.
- The final loop processes the 'country_data' (which contains information about players) and appends each player's data to the 'all_players' list.

```python
# Create a dataframe
columns = ["League", "Season", "Country", "Player", "Position", "Club", "Debut Date", "Appearances", "Goals"
df = pd.DataFrame(all_players, columns=columns)

# Save the dataframe to a CSV file
df.to_csv("players.csv", index=False)
display(df)
```

Figure 9: Creation of a dataframe and saving it into a csv file.

Lastly, from Figure 9, we create a dataframe from the list 'all_players' by adding the name of the columns and then save it in a csv file named 'players.csv'. The representation for the dataframe follows.

	League	Season	Country	Player	Position	Club	Debut Date	Appearances	Goals
0	premier-league	21/22	Morocco	Romain Saïss	Centre-Back	Wolverhampton Wanderers	22/09/2018	31	2
1	premier-league	21/22	Morocco	Hakim Ziyech	Right Winger	Chelsea FC	17/10/2020	23	4
2	premier-league	21/22	Morocco	Imrân Louza	Central Midfield	Watford FC	21/08/2021	20	0
3	premier-league	21/22	Morocco	Adam Masina	Centre-Back	Watford FC	15/09/2018	15	0
4	premier-league	21/22	Senegal	Sadio Mané	Left Winger	Liverpool FC	27/09/2014	34	16
...	...	...	...	...	...	...	...	...	...
1032	laliga	23/24	Ghana	Joseph Aidoo	Centre-Back	Celta de Vigo	24/08/2019	6	0
1033	laliga	23/24	Guinea	Sory Kaba	Centre-Forward	UD Las Palmas	25/08/2023	17	1
1034	laliga	23/24	Guinea	Mouctar Diakhaby	Centre-Back	Valencia CF	20/08/2018	14	1
1035	laliga	23/24	Guinea	Ilaix Moriba	Central Midfield	Getafe CF	13/02/2021	14	0

Figure 10: Players dataframe obtained through scraping from Transfermakt.

4 Data Storage

After scraping, it is crucial to save the data in a repository. Since we obtained tables from the two sources, we use a relational database management system (RDBMS), mySQL. In the RDBMS, a relational model is implemented, a logical model. It is a way of organizing data in a database based on the idea of representing data as rows in tables, each table having a unique name and a set of columns each with a unique name. Tables can be related to each other using foreign keys, which are columns that contain values that refer to the primary key of another table. This allows the data in the tables to be connected and accessed in a logical and structured way.

```python
# Library import
from sqlalchemy import create_engine
# Connection to the SQLite database
engine = create_engine('sqlite:///my_database.db')
# Save first dataframe in the mySQL database
df1.to_sql('table1', con=engine, if_exists='replace', index=False)
# Save second dataframe in the database
df2.to_sql('table2', con=engine, if_exists='replace', index=False)
```

Figure 11: Data storage in a relational database.

Data storage code explanation:

- From 'sqlalchemy' import 'create_engine', this line imports the 'create_engine' function from the 'sqlalchemy' library, which is a toolkit and ORM (Object Relational Mapper) for working with databases in Python. 'create_engine' is specifically used to set up connections to various types of database, like SQLite in this case.

- 'sqlite:///my_database.db', is a URL that tells SQLAlchemy to use SQLite as the database engine. The '///' part refers to the file path where the SQLite database (my_database.db) will be stored. Since this is a relative path (no leading /), the database file will be created in the current working directory. 'engine' now represents a connection to the SQLite database, which will be used to interact with the database later in the code.

- 'df1.to_sql('table1', con=engine, if_exists='replace', index=False)' This is a pandas method used to save a dataframe (df1) to an SQL database. The same procedure applies for the second dataframe.

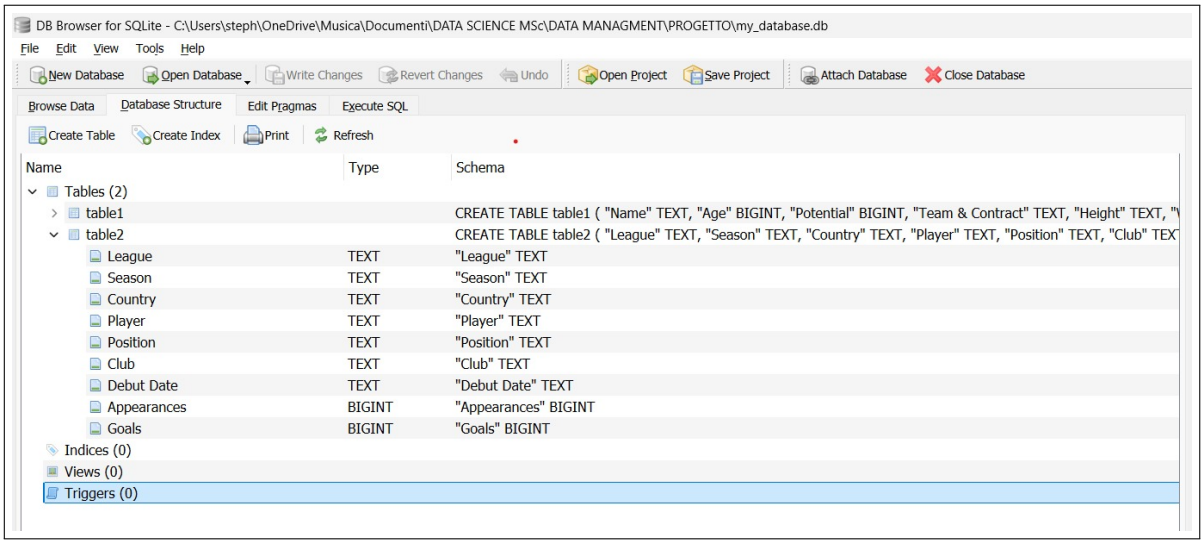


Figure 12: Graphical interface of mySQL containing the two tables in the relational model.

5 Data profiling

In the context of data management, data profiling refers to the process of examining, analyzing, and assessing the quality, structure, and content of data after it has been scraped but before it is integrated into a target system. The goal is to gain a deep understanding of the data's characteristics so that they can be cleaned, transformed, and integrated properly. It helps ensure that the data meet the required quality standards and are suitable for processing. Some activities involved in data profiling are:

- **Data Cleaning**: identify and address issues such as duplicates, missing values, incorrect formats, and outliers, before integration.

- **Data Transformation**: understanding the structure and quality of data to know how it should be transformed to fit integration, like creation of new features through other columns.

5.1 Data Profiling: Sofifa

Since we obtained 3 different tables for each season, we proceed by concatenating the data into one unique dataframe.

Follow the code:

```
csv_files=["player_2022final.csv","player_2023final.csv","player_2024final.csv"]
merged_data = pd.concat([pd.read_csv(file) for file in csv_files])

# Save the merged data into a single CSV file
merged_data.to_csv("players_all_3_seasons.csv", index=False)
```

Figure 13: Concat of the three different seasons scraped from Sofifa using selenium.

Pre-processing done on the dataset after concat:

- Since the variable 'Player_names' contains both the player name and role, we separate them to have two different columns for each.
- We divide the team and contract column to obtain two distinct columns for **completeness**.
- We then convert value (string to integer), wage (string to integer), height (feet to cm) and weight (libbras to kg) in the desired unit measure.
- We standardize and replace player positions based on the following roles dictionary for **consistency**:

Abbreviation	Position	Abbreviation	Position
GK	Goalkeeper	CM	Central Midfield
LB	Left-Back	RCM	Central Midfield
LWB	Left-Back	RM	Right Midfield
LCB	Centre-Back	LAM	Attacking Midfield
CB	Centre-Back	CAM	Attacking Midfield
RCB	Centre-Back	RAM	Attacking Midfield
RB	Right-Back	LW	Left Winger
RWB	Right-Back	LF	Forward
LDM	Defensive Midfield	CF	Forward
CDM	Defensive Midfield	RF	Forward
RDM	Defensive Midfield	LS	Forward
LM	Left Midfield	ST	Forward
LCM	Central Midfield	RW	Right Winger

Table 2: Football Player Positions

- To ensure correspondency with Transfermakt data, we normalize player names by lowercase and remove accents, and create a team name mapping dictionary for correspondency with names from Transefermakt, also for consistency in data.
- Lastly, we create an ID for each player by removing the point in the name of the players and create concatanating first letter of the name, team and season.

5.2 Data Profiling: Transfermakt

We check the missing values and consider how to manage them to solve any problem of completeness:

Column	Missing Percentage
League	0.00
Season	0.00
Country	0.00
Player	0.00
Position	0.00
Club	3.66
Debut Date	5.88
Appearances	0.00
Goals	0.00

Table 3: Missing Values Percentage for each column from Transfermakt dataframe.

From Table (3) the only variables with missings are 'Club' and 'Debut Date'. We will proceed in managing just 'Club' since players have not debuted with their clubs, so missing values in 'Debut Date' are fulfilled with 'Nas not debuted'.

Imputing missing clubs

```
In [2]: df = pd.read_csv('players.csv')
        print(df.info())
        missing_club = df[df.iloc[:, 5].isnull()]
        pd.set_option('display.max_rows', None)
        display(missing_club)
        pd.reset_option('display.max_rows')

        # Create a list with missing clubs
        clubs = ['Nottingham Forest', 'Fulham FC', 'AS Saint-Étienne', 'OGC Nice', 'OGC Nice', 'FC Girondins Bordeau
                 'ESTAC Troyes', 'AS Monaco', 'Olympique Marseille', 'FC Lorient', 'Stade Rennais FC', 'AJ Auxerre',
                 'AC Ajaccio', 'OGC Nice', 'Stade Rennais FC', 'FC Nantes', 'Stade Rennais FC', 'AC Ajaccio', 'Montp
                 'Stade Brestois 29', 'FC Lorient', 'Olympique Marseille', 'Stade Brestois 29', 'Stade Rennais FC',
                 'Torino FC', 'Atalanta BC', 'US Lecce', 'US Cremonese', 'FC Empoli', 'Torino FC', 'Frosinone Calcio
                 'Rayo Vallecano', 'Getafe CF', 'Real Sociedad', 'Real Betis Balompié']

        # Impute the missing clubs
        df.loc[df['Club'].isna(), 'Club'] = list(iter(clubs))
        print(df.info())
```

Figure 14: Manage missings from Transfermakt table.

The code, in figure (14), starts by defining a list of club names which will be used to replace missing (NaN) values in the 'Club' column of the dataframe. The missing values in 'Club' are replaced with values from the clubs list in a sequential manner using iter() and list().
Then, we look for any differences in structure, format, and context. This can include differences in the types of data (e.g. numeric, categorical, text), the names of the data attributes (e.g. 'Name', 'Player', 'Player name') and the definitions of the data attributes (e.g. 'Player' might refer to a unique identifier in one dataset and to a player name in another). The conflicts in the attributes found were: the player name. In the first source, Sofifa, we had the player name written as the first letter of the name and then the last name in full. In the second source, Transfermakt, both name and surname were written full. The solution adopted was to apocopate the name in the second source to obtain a correspondence in the players name.
The following figure (15) describes the preprocessing performed on the dataset obtained from Transfermakt, to ensure coherence with the one obtained from Sofifa:

```python
# Convert 'Name' and 'Club' to lowercase and remove accents
df['Player'] = df['Player'].apply(lambda x: unidecode(x.lower().replace("\xa0", " ")) if isinstance(x, str)
df['Club'] = df['Club'].apply(lambda x: unidecode(x.lower().replace("\xa0", " ")) if isinstance(x, str) else

# Function to abbreviate first name
def abbreviate_name(name):
    parts = name.split()
    if len(parts) > 1:
        return f"{parts[0][0]}. {' '.join(parts[1:])}"
    return name

# Apply function to the column ("Player") with the full name
df['Short Name'] = df['Player'].apply(abbreviate_name)

# Create the ID column
df["ID"] = df["Short Name"].str.replace(".", "") + "_" + df["Club"].str.replace(".", " ") + "_" + df["Seasor
df["ID"] = df["ID"].str.replace(" ", "_")

# Adjusting confuding positions
display(df['Position'].unique())
df['Position'] = df['Position'].replace({'Centre-Forward':'Forward', 'Second Striker':'Forward'})
display(df['Position'].unique())

display(df)
```

Figure 15: Pre-process of features from Transfermakt before integration.

- We standardize the features 'Name' and 'Club' to lowercase and remove accents for correspondence and consistency with the other dataset.
- With the function 'abbreviate_name(name)' we pass the player's name to abbreviate it with just the initial alphabet.
- Once we have finished with previous operations, for each player, we obtain an ID by joining the player's first letter of the name, club, and season.
- We then adjust the striker's position making 'Centre-Forward' and 'Second Striker' both 'Forward' for corrispondence in Sofifa's dataset.

Row	Season	Player	Club	...	ID
0	21/22	Romain Saiss	Wolverhampton Wanderers	...	r_saiss_wolverhampton_wanderers_21_22
1	21/22	Hakim Ziyech	Chelsea FC	...	h_ziyech_chelsea_fc_21_22
2	21/22	Imran Louza	Watford FC	...	i_louza_watford_fc_21_22
3	21/22	Adam Masina	Watford FC	...	a_masina_watford_fc_21_22
4	21/22	Sadio Mane	Liverpool FC	...	s_mane_liverpool_fc_21_22
⋮	⋮	⋮	⋮	⋮	⋮
1032	23/24	Joseph Aidoo	Celta de Vigo	...	j_aidoo_celta_de_vigo_23_24
1033	23/24	Sory Kaba	UD Las Palmas	...	s_kaba_ud_las_palmas_23_24
1034	23/24	Mouctar Diakhaby	Valencia CF	...	m_diakhaby_valencia_cf_23_24
1035	23/24	Ilaix Moriba	Getafe CF	...	i_moriba_getafe_cf_23_24
1036	23/24	Selu Diallo	Deportivo Alaves	...	s_diallo_deportivo_alaves_23_24

Table 4: Transfermakt Player Data after Pre-Processing

6 Data integration and enrichment

6.1 Data integration

In this phase, we combine the data obtained and profiled into a single unified view and load it into a central repository. The goal of data integration is to make the data easier to analyze and understand by bringing

together all relevant information in one place. The integration phase involves merging datasets from multiple sources, specifically Sofifa and Transfermarkt, to create a unified and comprehensive player profile. Initially, it was necessary to locate and extract relevant attributes from each source, such as physical statistics, skill ratings, and performance metrics from Sofifa, and match history, appearances, and market values from Transfermarkt. To align and merge these datasets, common identifiers such as player names, clubs, positions, and seasons were used. Additional processing was carried out to resolve inconsistencies in naming conventions and data formats between sources. In addiction, a unique identifier (ID) was created in the two datasets and used to merge both datasets. This step allowed for a standardized output structure, resulting in a single dataset that contains both technical performance and market-related data. This integrated dataset enables the creation of complete player profiles combining performance analytics with financial and contractual insights essential to support accurate and efficient scouting decisions.

```
df_merged = pd.merge(df1, df2, on='ID')
df_merged.to_csv('merged_dataset.csv', index=False)
```

Figure 16: Code for integration

This code uses the pandas library to merge the dataframes, based on the players' id. The pd.merge(df1, df2, on='ID') function performs an inner join by default, meaning it includes only the rows where the 'ID' values match in both DataFrames. The result of this merge is stored in a new dataframe called df_merged. The second line, df_merged.to_csv('merged_dataset.csv', index=False), saves the resulting dataset to a csv file.

6.2 Data enrichment

We rebuilt the famous fifa card that has a representative value for the most important features for football players: Speed, Shooting, Passing, Dribbling, Defending, Physical (goalkeepers have a personalized card with different features). The values of the outfield player cards are constructed as a linear combination of various features available in the database:

- Speed = $0.5 \cdot \text{Acceleration} + 0.5 \cdot \text{Sprint Speed}$
- Shooting = $0.2 \cdot \text{Finishing} + 0.05 \cdot \text{Heading Accuracy} + 0.1 \cdot \text{Volleys} + 0.15 \cdot \text{Curve} + 0.05 \cdot \text{FK Accuracy} + 0.15 \cdot \text{Shot Power} + 0.15 \cdot \text{Long Shots} + 0.15 \cdot \text{Penalties}$
- Passing = $0.1 \cdot \text{Heading Accuracy} + 0.3 \cdot \text{Short Passing} + 0.3 \cdot \text{Long Passing} + 0.3 \cdot \text{Vision}$
- Dribbling = $0.3 \cdot \text{Dribbling} + 0.3 \cdot \text{Ball Control} + 0.2 \cdot \text{Agility} + 0.1 \cdot \text{Reactions} + 0.1 \cdot \text{Balance}$
- Defending = $0.2 \cdot \text{Aggression} + 0.25 \cdot \text{Interception} + 0.3 \cdot \text{Defensive Awareness} + 0.25 \cdot \text{Standing Tackle}$
- Physical = $0.3 \cdot \text{Jumping} + 0.3 \cdot \text{Stamina} + 0.4 \cdot \text{Strength}$

While goalkeepers card is built considering the values of the columns GK Diving, GK Handling, GK Kicking, GK Reflects, Gk Positioning, plus the Speed built in the same as for the outfield players. We also create two variables 'Main' and 'Alternative' position for players who could play in different positions.

7 Data quality assessment

7.1 Completeness

The completeness of a dataset is the coverage with which the observed phenomenon is represented in the dataset. Data not being available in one source made the data incomplete. For example: Rayan Aït Nouri had French citizenship in the 22/23 season and acquired Algerian citizenship in the 23/24 season. As well as, Ademola Lookman who had English citizenship in the 21/22 season and acquired the nigerian citizenship in the 22/23 season.

7.2 Consistency

For consistency in the data, these problems were fixed:

1. Different player names across sites, which caused ID mismatch due to inconsistent naming. For example: Pape Matar Sarr has the 'Short Name' as 'p. matar sarr' from one source and 'p. sarr' from another. Other example: Hamed Junior Traorè has the 'Short Name' as 'h. junior traore' in one and 'h. traore' in the other.
2. Special characters in surnames on Transfermarkt but not on Sofifa.
3. Differences due to relegation and promotion.
4. Some players transferred mid-season (winter market) had two clubs assigned, the previous club and the new club. These were fixed manually.

7.3 Currency

Some data were outdated or were not aligned with the expected time frame. For example, some players changed teams midseason, but the data do not reflect this evolution in a timely manner. Also, the scraping was done at the end of the three seasons we considered, with all data already available.

7.4 Accuracy

Since we are not in a model prediction task, accuracy in this case refers to data correctness, consistency, and completeness after integrating multiple data sources. These issues have been explained and solved in the previous steps.

8 Final Data Explorative Analysis

To analyze the distribution of players across both leagues and countries during the seasons 2021/2022, 2022/2023, and 2023/2024, we performed SQL queries on the merged dataset. The purpose was to generate two summary tables: one grouping players by league and season, and the other by country and season

The first query was used to count the number of players from each African country for every season:

```
SELECT
  Country,
  Season_x,
  COUNT(*) AS num_giocatori
FROM
  merged
GROUP BY
  Country, Season_x
ORDER BY
  Country, Season_x;
```

This query aggregates data by country and season, allowing us to assess how the presence of players from specific nations such as Senegal, Nigeria, or Ivory Coast has changed over time. The resulting table supports comparative and temporal analyses and highlights key exporting nations in African football. The use of '`GROUP BY`' ensures aggregation at the appropriate level of detail, while '`ORDER BY`' structures the output for interpretability.

Country	21/22	22/23	23/24
Algeria	20	15	20
Burkina Faso	3	5	5
Cameroon	15	14	15
Capo Verde	-	3	2
Democratic Republic of Congo	6	7	5
Egypt	3	4	4
Ghana	21	22	20
Guinea	6	7	12
Ivory Coast	28	25	31
Mali	18	21	20
Marocco	20	22	31
Nigeria	23	19	19
Senegal	32	28	35
South Africa	-	1	2
Tunisia	7	7	7

Table 5: Number of players by country and season

The second query was formulated to evaluate the distribution of these players across the five major European leagues:

```
SELECT
  League,
  Season_x,
  COUNT(*) AS num_giocatori
FROM
  merged
GROUP BY
  League, Season_x
ORDER BY
  League, Season_x;
```

This query groups players by league and season, allowing us to observe the dynamics of league-specific recruitment. For example, it highlights the consistent prominence of Ligue 1 in hosting African talent, in contrast to relatively lower but growing numbers in the Premier League or La Liga. These patterns provide valuable information on scouting behaviors and migration channels within elite European football.

League	21/22	22/23	23/24
Bundesliga	19	22	23
La Liga	16	26	25
Ligue 1	97	80	101
Premier League	39	39	47
Serie A	31	33	32

Table 6: Number of players by league and season

Taking into account the data related to the number of African players in the top five European football leagues (Bundesliga, La Liga, Ligue 1, Premier League, and Serie A) over three seasons (2021/22 to 2023/24), some noteworthy patterns and dynamics emerge.

With regard to the country of origin of the players, it can be observed that Senegal, Ivory Coast, Nigeria, and Morocco consistently exhibit the highest levels of representation. On average, Senegal contributes more than 30 players per season, reaching a peak of 35 in the 2023/24 season. Morocco's contribution shows a marked increase, from 20 in 2021/22 to 31 in 2023/24. A similar trend is seen for Ivory Coast, with a progression from 28 to 31 players. In contrast, Nigeria experiences a moderate decline, from 23 to 19 players, a figure that remains stable in the last two seasons.

The range of variation among countries is quite wide. Some countries, such as Capo Verde and South Africa, appear only in the final two seasons, with a maximum of 3 and 2 players, respectively. The presence of these nations, while limited, may indicate the beginning of a broader process of internationalization in their football development systems.

In addition to the larger countries, other nations such as Guinea and Mali show moderate but significant increases. For example, Guinea nearly doubles its representation, from 6 in 2021/22 to 12 in 2023/24. Egypt and Tunisia remain stable, each contributing between 3 and 7 players across all three seasons. The data do not show any country dropping out completely, which confirms the continuity of African player exports.

In terms of league-wise analysis, the highest concentration of African players is consistently observed in Ligue 1 (France), with a particularly high value in the 2023/24 season (101 players). This may be attributed to the linguistic and historical ties between France and numerous West and North African countries. The second most represented league is the Premier League, which, while remaining stable in the first two seasons (39 players), rises to 47 in 2023/24. Both La Liga and the Bundesliga demonstrate a slight but constant increase over time, suggesting growing interest in African talent in Spanish and German clubs. Serie A remains stable throughout, oscillating between 31 and 33 players.

The overall distribution of players by league also exhibits moderate variability, but remains relatively concentrated. Most African players are absorbed by Ligue 1, while the remaining leagues share the rest in more balanced proportions. The increase in 2023/24 in almost all leagues indicates a general trend of increasing demand for African players, possibly linked to improved training systems, international visibility (e.g., AFCON and World Cup performances), and the success of youth academies across the continent.

In conclusion, the dataset shows that the representation of African players in the top five European leagues is not only significant, but also evolving. Some countries (notably Senegal, Morocco, and Ivory Coast) appear structurally established within these competitions, while others are in the early stages of integration. The trends are mostly upward, both at the national and league levels, confirming the growing importance of African talent in European football.

Correlation between the number of players and the FIFA ranking points

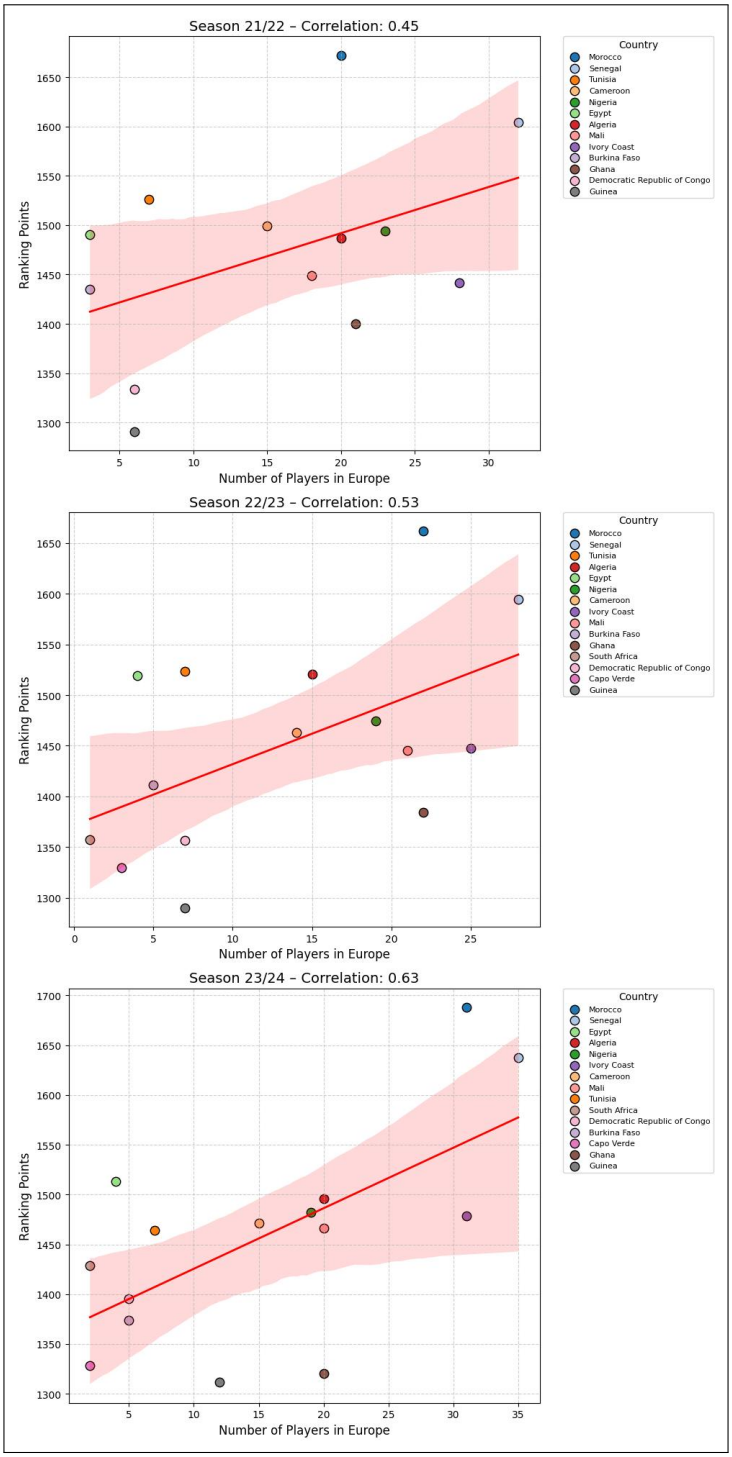


Figure 17: Correlation between the number of players playing in the top five European leagues and the FIFA ranking points of the nation for each year.

To better understand the relationship between the presence of African players in top-tier European football and their respective nations' performance on the international stage, we calculated the Pearson correlation coefficient between the number of players per country in the top five European leagues and their corresponding FIFA ranking points for each of the three seasons under analysis: 2021/2022, 2022/2023, and 2023/2024.

Seasonal Correlation Coefficients:

- **2021/2022:** $r = 0.45$
- **2022/2023:** $r = 0.53$
- **2023/2024:** $r = 0.63$

These coefficients indicate a moderate and progressively strengthening positive correlation over the 3-year period. Although the relationship is not exceptionally strong, it becomes increasingly evident that nations with more players in the top five leagues tend to achieve better results in the FIFA rankings, thus in international competition.

- In **2021/2022**, the correlation is relatively low ($r = 0.45$), suggesting that player presence in elite European leagues was not a dominant factor in influencing FIFA ranking points at that stage. This could be due to factors such as strength in the domestic league, coaching strategies, or transitional phases in national squads.
- In **2022/2023**, the correlation strengthens ($r = 0.53$), reflecting a growing alignment between international performance and the professional environment in which players develop.
- By **2023/2024**, the correlation increases further to $r = 0.63$. This moderate to strong relationship indicates that teams increasingly benefit from having players competing at the highest club level in Europe, likely due to higher match intensity, better tactical preparation, and exposure to elite competition.

Analyzing the correlation in countries performances:

- Senegal and Morocco consistently show high player presence and strong FIFA ranking points, reinforcing their elite status in African football.
- Ivory Coast, Ghana and Nigeria have strong player representation, but occasionally show less-than-expected FIFA ranking scores, suggesting possible inefficiencies in squad management or tournament outcomes.
- However, countries like South Africa or Burkina Faso show low player presence in top leagues but maintain decent FIFA points, possibly due to strong cohesion, domestic league performance, or favorable regional tournament results.
- Guinea and Mali appear as middle cases gradually improving both in player representation and international performance, showing potential for further rise.

The analysis shows a clear upward trend in the correlation between player exports to elite European clubs and the international success of African nations. Although correlation values remain moderate, the steady increase highlights that the development and performance of national teams are increasingly tied to the quality of club environments in which their players are involved. These insights could help federations and scouts identify strategic directions for player development and recruitment, potentially aiming to increase their representation in competitive leagues as a pathway to enhance national team performance.

Top African Players by Season

The following tables present the best African players in consecutive seasons: 2021/2022, 2022/2023, and 2023/2024. For each season, the tables list players based on their overall ratings (a weighted mean) and prominence within the top European leagues.

Each entry includes the player's name, nationality, primary playing position, overall rating (as per game data or scouting reports), market value in millions of euros, and weekly wage in thousands of euros. This data provides insights into both the on-field performance and economic valuation of Africa's elite football talents.

Several players demonstrated remarkable consistency throughout all three seasons. Notably, Mohamed Salah from Egypt and Achraf Hakimi from Morocco maintained top tier performance levels with overall ratings of 89 and 91-92 respectively, reinforcing their positions as Africa's superstars. Thomas Partey from Ghana and Ismael Bennacer from Algeria also maintained high performance levels, with Bennacer howling a notable improvement from an overall rating of 86 in 2021/2022 to 91 by 2023/2024.

Player	Country	Position	Overall	Market Value (M)	Weekly Wage (K)
Edouard Mendy	Senegal	Goalkeeper	82	30	105
Hassane Kamara	Ivory Coast	Left-Back	83	11	30
Ramy Bensebaini	Algeria	Centre-Back	82	21	31
Edmond Tapsoba	Burkina Faso	Centre-Back	81	48.5	49
Achraf Hakimi	Morocco	Right-Back	91	69.5	100
Franck Kessie	Ivory Coast	Central Midfield	87	49.5	51
Thomas Partey	Ghana	Central Midfield	87	36.5	115
Ismael Bennacer	Algeria	Central Midfield	86	30	41
Sadio Mane	Senegal	Left Winger	89	101	270
Andy Delort	Algeria	Forward	83	25.5	49
Mohamed Salah	Egypt	Right Winger	89	101	270

Table 7: Top players for season 2021/2022

Player	Country	Position	Overall	Market Value (M)	Weekly Wage (K)
Edouard Mendy	Senegal	Goalkeeper	84	51	130
Fode Ballo-Toure	Senegal	Left-Back	78	3	32
Ramy Bensebaini	Algeria	Centre-Back	82	15	28
Edmond Tapsoba	Burkina Faso	Centre-Back	80	35	49
Achraf Hakimi	Morocco	Right-Back	91	53.5	90
Franck Kessie	Ivory Coast	Central Midfield	90	47.5	165
Ismael Bennacer	Algeria	Central Midfield	89	40	63
Thomas Partey	Ghana	Central Midfield	86	36	115
Krepin Diatta	Senegal	Left Midfield	81	8.5	34
Sadio Mane	Senegal	Forward	88	99.5	145
Mohamed Salah	Egypt	Right Winger	89	115.5	270

Table 8: Top players for season 2022/2023

Player	Country	Position	Overall	Market Value (M)	Weekly Wage (K)
Andre Onana	Cameroon	Goalkeeper	85	46	130
Issiaga Sylla	Guinea	Left-Back	77	2.7	18
Ramy Bensebaini	Algeria	Centre-Back	86	19	50
Massadio Haidara	Mali	Centre-Back	81	6.5	30
Achraf Hakimi	Morocco	Right-Back	92	50.5	90
Ismael Bennacer	Algeria	Attacking Midfield	91	53.5	84
Thomas Partey	Ghana	Central Midfield	86	45	150
Ibrahim Sangare	Ivory Coast	Central Midfield	85	31	70
Hassane Kamara	Ivory Coast	Left Midfield	81	3.7	15
Inaki Williams	Ghana	Forward	85	25.5	37
Mohamed Salah	Egypt	Right Winger	89	85.5	260

Table 9: Top players for season 2023/2024

Market values broadly reflected these performance trends. For example, Bennacer's market value nearly doubled over the period, increasing from €30M to €53.5M, highlighting his rising profile in European football. In contrast, Achraf Hakimi's market value declined slightly despite consistent performance, possibly due to external factors such as contract age or transfer market volatility.

In terms of weekly wages, Mohamed Salah remained the highest-paid player across all three seasons, earning €270K per week. Franck Kessie from Ivory Coast saw a dramatic wage increase from €51K in 2021/2022 to €165K in 2022/2023, signaling a major contract renegotiation or high-profile transfer. Ismael Bennacer again stood out with a substantial wage increase from €41K to €84K over the analysis period, further underscoring his upward trajectory.

From a geographical point of view, Algeria, Senegal, the Ivory Coast, and Ghana emerged as the countries most represented, collectively accounting for the majority of top players throughout the seasons. Algeria, in particular, contributed key players in both defensive (Ramy Bensebaini), midfield (Ismael Bennacer) and attacking (Andy Delort) roles. Senegal showed breadth with representation in goalkeeping (Edouard Mendy), midfield (Krepin Diatta), and attacking positions (Sadio Mané).

9 Conclusions and future developments

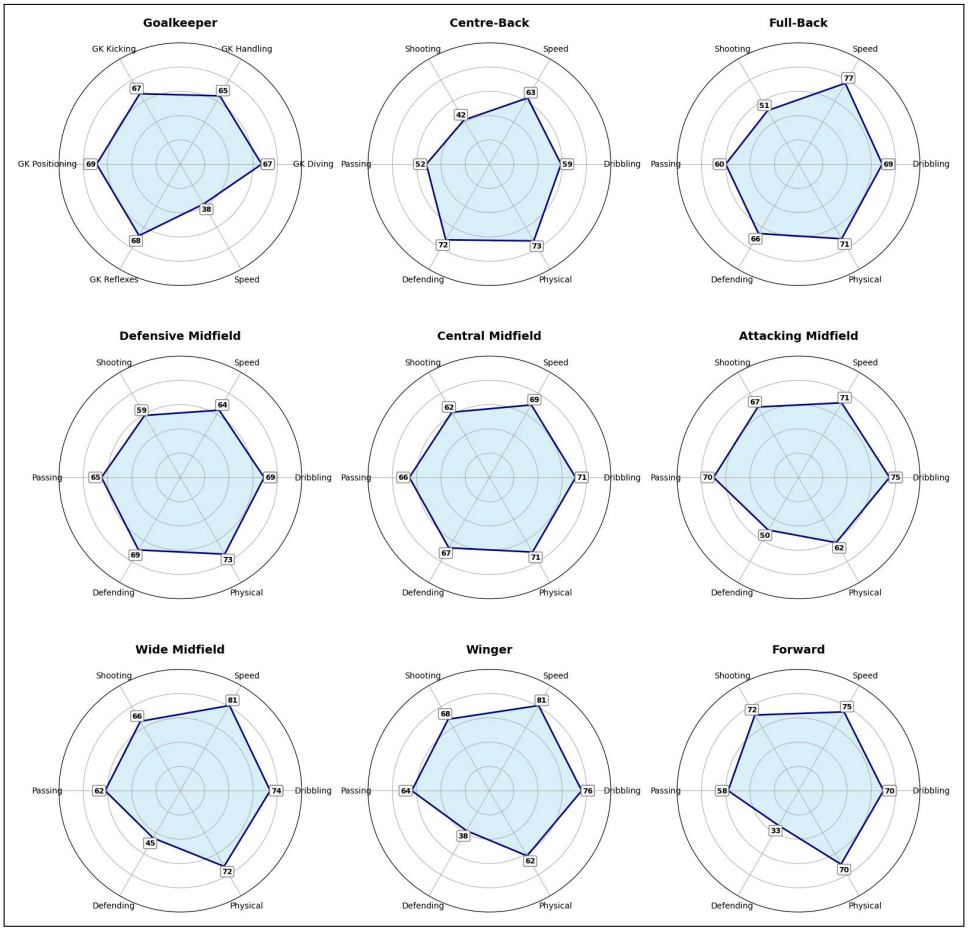


Figure 18: Radar charts that show the characteristics of players for each position

Strikers and Wingers (LW, ST, RW): Players in attacking positions, particularly strikers and wingers, exhibit the most extreme specialization among all roles. Their profiles are characterized by high speed, shooting

and dribbling qualities, which reflect their role in penetrating defensive lines, finishing chances, and engaging in 1-vs-1 situations. The radar charts show that speed is typically one of the highest dimensions for wingers, who often rely on explosive acceleration and sprinting to create space. Shooting is more prominent among central strikers. In contrast, passing is relatively a minor characteristic for these roles, especially for pure finishers. These patterns suggest that clubs looking to recruit attacking players often prioritize direct offensive contributions over defensive work or playmaking ability.

Midfielders (CDM, CM, CAM): Midfielders present a more balanced radar profile compared to attackers. However, distinct sub-profiles emerge depending on the specific role:

- **CDM (Defensive Midfielders)**: They excel in defending and physical structure, supporting their task of breaking opposition play and protecting the backline. Interestingly, they also show decent passing skills, which underlines the modern need for defensive midfielders to be effective distributors under pressure.
- **CM (Central Midfielders)** The chart displays relatively good statistics in most attributes, none particularly low. This suggests that versatility and consistency are critical characteristics in this position.
- **CAM (Attacking Midfielders)** They tend to score highly in passing and dribbling, making them key in transition play and chance creation. Their shooting scores are moderate, as they often arrive in the box late or attempt shots from distance.

This heterogeneity makes midfield roles particularly complex to evaluate. Radar analysis enables clubs to differentiate between creative and destructive midfielders and to identify players who can operate effectively in hybrid roles.

Defenders (LB, CB, RB): The profiles of defenders, especially central defenders (CB), are dominated by high defensive and physical values. These metrics correspond to key attributes such as tackling, marking, interception, and aerial strength. Speed is generally high for fulbacks (LB, RB), who often engage in overlapping runs, but lower for centerbacks. Shooting is the lowest for all defender profiles, reflecting their limited involvement in offensive finishing. Notably, some fullbacks show decent passing and dribbling values, suggesting an increased tactical emphasis on wide defenders' ability to contribute to ball progression and crossing.

Summary and Implications: This positional analysis reveals how the technical and athletic attributes of the players vary systematically with their role on the field. It also highlights the specificity required in scouting processes: a high-quality player is not simply one with high average values but one whose attribute profile aligns with the tactical and positional demands of their role. Therefore, teams aiming to recruit efficiently from the pool of African players in the top five European leagues should focus on position-specific radar benchmarks. These profiles can also help identify 'under the radar' players who may not have outstanding global ratings but match the ideal template for a given position.

Player	Age	Country	Main Position	Preferred Foot	Overall	Market Value (M)
Achraf Hakimi	22	Morocco	Right-Back	Right	91	69.5
Krepin Diatta	22	Senegal	Right Midfield	Right	82	15
Hicham Boudaoui	21	Algeria	Central Midfield	Right	82	11.5
Cheick Doucoure	21	Mali	Central Midfield	Right	78	8.5
Edmond Tapsoba	22	Burkina Faso	Centre-Back	Right	81	48.5
Victor Osimhen	22	Nigeria	Forward	Right	80	43
Bryan Mbeumo	22	Cameroon	Forward	Left	79	9
Cheick Doucoure	22	Mali	Central Midfield	Right	79	10
Hicham Boudaoui	22	Algeria	Central Midfield	Right	82	12
Mohamed Camara	22	Mali	Central Midfield	Right	83	22.5
Wilfried Singo	21	Ivory Coast	Right Midfield	Right	80	16
Hamed Junior Traore	22	Ivory Coast	Attacking Midfield	Right	82	31.5
Mohammed Kudus	22	Ghana	Attacking Midfield	Left	85	35.5
Wilfried Singo	22	Ivory Coast	Right Midfield	Right	80	16
Fares Chaibi	20	Algeria	Central Midfield	Right	79	12.5

Table 10: Player information for 21/22, 22/23, and 23/24 seasons.

The table highlights a group of promising African players who are emerging as key talents in the top five European football leagues. The table includes players from various positions, indicating the breadth of talent across different roles on the field. Already successful players like Achraf Hakimi (Morocco), who boasts an impressive 91 overall rating and a market value of €69.5M, exemplify the top-tier African talent currently playing in European leagues. The table also features Mohammed Kudus (Ghana), who has consistently grown in value and rating, reflecting his increasing prominence. Other players, such as Victor Osimhen (Nigeria) and Bryan Mbeumo (Cameroon), have shown strong performances with significant appearances and steady market value, further solidifying their place among Europe's rising stars.

The players in this table not only showcase potential but also demonstrate actual contributions through their appearances in matches. While some have established themselves as regular starters, others, such as Fares Chaibi (Algeria), are on the cusp of breaking out, making their presence felt with growing market values and appearances over the years. These young talents represent Africa's hidden gems, contributing to the vibrant football talent emerging from the continent to the European stage. Their progress over time reflects the increasing competitiveness of African players in top European leagues, making them exciting prospects for the future.

Roles and Personal Considerations:

- **Gianluca Licciardello**: Responsible for data scraping from Transfermarkt and overall data analysis.
- **Stephen Adu Poku Yeboah**: Responsible for data scraping from Transfermarkt and data storage and queries on MySQLite.
- **Emmanuel Ampah**: Responsible for data profiling and data quality and overall data analysis.

The main challenge we faced was the lack of matching IDs due to several factors, which influenced the quality of our analisis. Overall, we are satisfied with our project that we consider a reliable source for talent scouts interested in players from Africa. The lack of a unique source of data for scouting makes it difficult to get complete data, both in play and in video. So we agree that the creation of a unique webiste aimed at scouting would be the perfect solution for scouters to make their job virtually easier without the need to move around the world or to have excessive screen time. It would be very fast (they should not fly around the world or spend time scraping). In terms of costs, it would be more expensive (licenses prices and APIs are getting higher and higher), but it could generate bigger revenues. Of course, it should not substitute the traditional work of the talent scout, instead it should justapose it.

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